

Past 64 spins only tempering reaches the threshold; mixed grids stay out of reach

- Long chain
- Five cold chains
- Five replicas, $k=1$
- Five replicas, $k=4$
- Nine replicas, $k=1$
- Nine replicas, $k=4$

8x8 ferro (64 spins)

16x16 ferro (256 spins)

32x32 ferro (1024 spins)

8x8 mixed (64 spins)

16x16 mixed (256 spins)

32x32 mixed (1024 spins)

Mean edge MAE (3 seeds)

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Elapsed sweeps T (long runs 5T)

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Dotted line: qualification threshold 0.05, which must hold at every larger budget. Exact Kac-Ward references.
CPU software simulation, two-colour block Gibbs at beta 4, 16 trials per cell; no hardware claim.